MIL-C-43347D

26 September 1972

SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, POLYESTER, LOW AIR PERMEABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers polyester multifilament cloth.

1.2 Classification.- The cloth shall be of the following types as specified (see 6.2):

Type I - 8.5 Ounces
Type II - 10.5 Ounces

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics.

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections.
FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 8305

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Publications

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard sample.- The cloth shall be equal to or better than the standard sample (see 6.3) with respect to all characteristics for which the standard sample is referenced.

* 3.2 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.5).

3.3 Material.-

3.3.1 Yarns.-

3.3.1.1 Type I.- The yarns shall be polyethylene glycol teraphthalate, bright, multifilament. The warp shall be 220 denier (± 3 percent), 2 ply for the plain weave and 220 denier (± 3 percent), singles for the alternate weave (oxford). The filling shall be of 220 denier (± 3 percent), 3 ply. The yarns shall be tested as specified in 4.3.1.

3.3.1.2 Type II.- The yarns shall be polyethylene glycol teraphthalate, bright, multifilament. The warp shall be 440 denier (± 3 percent) singles and the filling shall be 1100 denier (± 3 percent) singles. The yarns shall be tested as specified in 4.3.1.

3.4 Nonfibrous materials.- Prior to the application of the water repellent treatment, the cloth shall not exceed 2.0 percent starch and protein content, including chloroform-soluble and water-soluble material when tested as specified in 4.4.

3.5 Color.- The color shall be natural.

3.6 Physical requirements.- The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

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TABLE I.- Physical requirements

Type	Weight oz/sq. yd.	Yarns per inch minimum		Breaking strength minimum Pounds		Tearing strength minimum Pounds		Air permeability cu ft/min/sq. ft. 1/
	Minimum	Warp	Filling	Warp	Filling	Warp	Filling	Maximum
I	8.5	64 2/	43	475	540	25	30	2.0
II	10.5	70	35	550	750	30	35	2.0

1/ At a pressure drop across the cloth of 6 inches of water.

2/ The minimum warp texture for the oxford weave shall be 128 (see 3.6.1.2).

3.6.1 Weave -

3.6.1.1 Type I and type II.- The weave for type I and type II shall be plain (see 4.4).

3.6.1.2 Type I alternate weave.- The alternate weave for type I may be woven with two warp ends weaving as one using a single 220 denier yarn. The minimum ends per inch shall then be 128. All other requirements for type I are applicable (see 4.4)

3.6.2 Width.- The finished cloth shall be a minimum of 41 inches in width, inclusive of selvages.

3.7 Finish for type I and II.- The cloth shall be scoured, high temperature heat set and calendered and water repellent treated with a silicone emulsion, to result in a finish equal to that of the standard sample and to meet the test requirements of this specification (see 4.4, 6.3.1 and 6.4).

3.7.1 Water repellency.- The silicone treated finished cloth shall show spray ratings no less than 90, 90, 90 when tested as specified in 4.4

* 3.8 Length and put-up - Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put up on full width rolls as specified in PPP-P-1133.

3.9 Fiber identification.- Each roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

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3.10 Face identification.- The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.11 Workmanship.- The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection.- When required (see 6.2), the preproduction sample submitted in accordance with 3.2 shall be visually examined for appearance, color and finish and shall be tested in accordance with methods specified in table II and 4.4.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition to the quality assurance provisions of the subsidiary specifications, testing shall be performed on components and materials listed in table II for the test characteristics shown.

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TABLE II.- Components

Characteristics	Requirement paragraph	Test methods
Material identification	3.3.1	1/ 1600 2/
Denier	3.3.1 and 3.6.1.?	1/ 4021 2/
Luster	3.3.1	1/
Ply		
Warp	3.3.1	1/
Filling	3.3.1	1/

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ Refers to FED-STD-191.

4.3.2 Examination of the end item.- Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.3.

* 4.3.2.1 Yard-by-yard examination.- Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 45.0 points. If one roll exceeds 45.0 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rol in the second sample exceeds 45.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

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4.3.2.1.1 Demerit points.- Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Objectionable odor
Width less than specified
Color other than natural
Overall uncleanness or discoloration
Character of finish not equal to that of the standard sample

4.3.2.2 Examination for length.-

4.3.2.2.1 Individual rolls.- During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.2.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.3.2.3 Examination for compliance with the Rules and Regulations Under the Textile Fiber Products Identification Act and face marking.- During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Rules and Regulations Under the Textile Fiber Products Identification Act and face marking. Each roll not labeled in accordance with this Act or with the face stamping missing from either or both ends, or stamped on the wrong side shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

* 4.3.3 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with the section 5 requirements.

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4.4 Tests.- The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods, except for air permeability and water repellency, for which they apply to individual determinations. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 yard. Except for air permeability (type II), the sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

For air permeability (type II), the sample size shall be in accordance with level S-3 of MIL-STD-105 and the acceptable quality level shall be 0.65 defects per 100 units (yards). The sample units for testing shall be as follows:

For nonfibrous material content - 1/4 yard full width of cloth prior to water repellent treatment.
For air permeability - 1/2 yard full width of finished cloth put-up on a tube.
For all other tests - 3 continuous yards full width of finished cloth.

TABLE III.- Methods of testing

Characteristics	Requirement paragraph	Test method
Unfinished cloth.		
Starch and protein content - including chloroform-soluble and water-soluble material	3.4	2611
Finished cloth.		
Weight	3.6	5041
Yarns per inch		
Warp	3.6	5050
Filling	3.6	5050

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TABLE III.- Methods of testing (cont'd)

Characteristics	Requirement paragraph	Test method
Finished cloth. (cont'd)		
Breaking strength		
Warp	3.6	5100
Filling	3.6	5100
Tearing strength		
Warp	3.6	5132
Filling	3.6	5132
Air permeability	3.6	5450 1/
Weave		
Type I and type II	3.6.1.1	Visual 2/
Type I alternate	3.6.1.2	Visual 2/
Finish	3.7	3/
Water repellency	3.7.1	5526

1/ Test method 5450 shall be used except that a High Pressure Differential Air Permeability Machine shall be used. This machine shall be equipped with a manometer capable of measuring a pressure drop across the cloth under test of 1 to 21 inches of water and this manometer shall be graduated to read pressure in inches of water. In addition, the pressure drop across the orifice indicated by the vertical manometer shall be not less than 3 inches (a pressure drop of less than 3 inches across the orifice as indicated by the vertical manometer shall be permitted provided that at the time of testing the smallest possible orifice is being used) nor more than 20 inches. The determination shall be made at a pressure drop across the cloth of 6.0 inches of water. A test failure shall be defined as 1 or more individual determinations failing to meet the specified requirement.

2/ One determination shall be made from each sample unit and the result reported as 'pass' or 'fail'.

3/ The supplier shall submit a certificate of compliance stating that the finished cloth has been scoured, high temperature heat set, calendered and treated with a silicone emulsion.

* 5 PREPARATION FOR DELIVERY

5.1 Put-up and packing - Put-up and packaging shall be level A or C as specified (see 6.2)

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5.1.1 Levels A and C.- The cloth shall be put up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use.-

6.1.1 Type I.- The type I cloth covered by this specification is intended for use as follows:

(a) The cloth for the slide fastener panel in Tent, Radome, Air Supported, Nike Hercules System, Single Wall 27 feet.
(b) The cloth for the lacing panel in Tent, Nike Hercules, Air Supported.

6.1.2 Type II.- The type II cloth covered by this specification is intended for use in the construction of the Tent, Radome, Air Supported.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of cloth required (see 1.2).
(c) When a first article sample is required (see 3.2, 4.2 and 6.5).
(d) Minimum length if other than specified (see 3.8).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.3.1 The standard sample for types I and II were finished as follows (see 3.7):

High alkaline scour at a boil.
Dry.
Silicone emulsion applied.
Slack heat set at 410° F.
High temperature calender 295°F. at pressure of 50 tons.

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6.4 Effect of heat.- It is to be noted that the polyester fibers available today vary in their shrinkage potential under the influence of heat. Based upon the supplier's choice of polyester yarn, appropriate time-temperature adjustments must be exercised (see 3.7).

* 6.5 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of 3-1/2 yards of finished cloth and 1/4 yard of cloth prior to the water repellent treatment. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.6 Marginal identification.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians.

Army - GL
Air Force - 11

Review activities:

Army - MI
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Preparing activity:

Army - GL

Project No. 8305-0877

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

Cloth, Plain Weave, Polyester, Low Air Permeability MIL-C-43347D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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